

NOTE: Next বুধবার lab mid
Mid Term examinationর
মিলেবার, marks - 20/30

String handling থেকে
একটা question থাকবে

Second CT date = ??

PDF from book

- ☐ Java Strings are immutable. `<< java.lang. String >>`
- ☐ String, StringBuffer, StringBuilder are FINAL
None of them can be subclassed.
- ☐ These three implement CharSequence interface.
- ☐ A string object cannot be changed
after creation.

```
psvm(...){  
    String s = "ABC";  
    s.concat("DEF");  
    System.out.println(s); // prints only "ABC"  
}
```

```
psvm(...){  
    String s = "ABC";  
    s = s.concat("DEF"); // assignment operation  
    System.out.println(s); // ABCDEF  
}
```

`"+" operator = concat()` // concatenation

String constructors

Default = new String()

Parameterized =

① Array char chars[] = {"a", "b", "c"};
String s = new String(chars);

② Range fixing

String(char chars[], int startIndex, int numChars)

PSVM (...) {

char chars[] = {"a", "b", "c", "d", "e", "f"};

String s = new String(chars, 2, 3);

System.out.println(s); // cde

String length

| s.length(); |

empty string length = 0

① ! empty string ≠ NULL

String Operations

① Concatenation { LTR operation }

i) s = "four" + 2 + 2 ⇒ four 22

ii) s = "four" + (2 + 2) ⇒ four 4

② toString

⇒ Return Type

always String হতে

i) public keyword use

হয়তো হতে (during method overriding)

ii) calls automatically

ii) Java built-in method

②

When toString() is called automatically?

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→ during ~~system.put~~.

system.out.println('')

→ during concatenation of string & objects.

```
class Box{
```

```
    int w, h;
```

```
    Box(int w, int h){
```

```
        this.w = w;
```

```
        this.h = h;
```

```
    }
```

```
    public String toString(){
```

```
        return "Dimension are: " + w + " by " + h + ".";
```

```
    }
```

```
}
```

```
public class Main{
```

```
    public static void main(String[] args){
```

```
        Box b = new Box(10, 5);
```

```
        String s = "Box " + b; // String s = b
```

```
        System.out.println(b);
```

```
        System.out.println(s);
```

```
    }
```

```
}
```

"toString() can be overridden in any class"

↑ output?

String operations

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③ Character Extractions

char charAt(int where) single character extraction

```
char s = "abc".charAt(1);  
System.out.println(s);    // b
```

multi character extraction

```
void getChars(int sourceStart,  
              int sourceEnd,  
              char target[],  
              int targetStart)
```

```
psvm(...) {
```

```
String s = "This is a demo of the getChars Method.";
```

```
int start = 10;
```

```
int end = 14;
```

```
char buf[] = new char[end - start];
```

```
s.getChars(start, end, buf, 0);
```

```
System.out.println(buf);
```

④ String comparison (boolean type)

→ boolean equals(Object str) // case sensitive

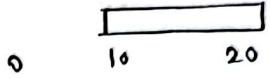
→ boolean equalsIgnoreCase(String str) // case insensitive

self study : equals vs ==

String Operations

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⑤ String Searching

→ `indexOf(String str, int startIndex)` → 

→ `lastIndexOf(String str, int startIndex)` 

`indexOf(t) = 7`

`lastIndexOf(t) = 54 65`

0 → Now 3 is 6 7 10 11 15 19 23 28 32
to 35 come 40 to 43 the 47 aid 51 of 54 55 their 60 ~~camp~~ 65
country ← 67

`indexOf(the) = 7`

`lastIndexOf(the) = 55`

`indexOf(t, 10) = 11`

`lastIndexOf(t, 60) = 7`

`indexOf(the, 10) =`

`lastIndexOf(the, 60) = 55`

⑥ Modifying a String

→ `substring(int startIndex, int endIndex)`

→ `concat`

→ `replace(char original, char replacement)` // all occurrences

→ `trim` // remove whitespaces

~~→ `replaceAll`~~

String operations

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⑦ changing cases

- ① toLowerCase();
- ② toUpperCase();